

Annuity Liability Valuation and Sensitivity Analysis

Human Mortality Database — UK Gilt Yield Curves — Python/Excel

This report builds a simple actuarial valuation framework for a single-life annuity liability. The model projects survival-weighted pension and expense cashflows using UK period life table data, discounts the expected cashflows using a Bank of England nominal gilt spot curve, and tests the impact of interest-rate, longevity, mortality, inflation and expense assumptions on the best-estimate liability (BEL).

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1 Executive summary

This project values a simplified single-life immediate annuity liability using survival-weighted cash-flows and a UK nominal gilt spot curve. The mortality basis is the Human Mortality Database UK total both-sex period life table, while discounting uses the Bank of England nominal gilt spot curve.

The base annuity assumes a 65-year-old annuitant receiving annual payments of £12,000, with annual expenses of £75 and pension increases modelled using a constant 2.5% inflation assumption. Payments are assumed to be made annually in arrears. The latest available HMD year in the model run was 2022, and the Bank of England curve date used was 2026-05-21.

The base best-estimate liability was £168,353.85. The sensitivity analysis showed that the liability is most exposed to discount-rate and inflation assumptions. A 100bps decrease in rates increased the BEL by £17,647, while a 100bps increase in inflation increased the BEL by £16,181. A 10% reduction in mortality rates, used as a simple longevity improvement stress, increased the liability by £5,060.

Overall, the project demonstrates the main mechanics of annuity liability valuation: mortality-based survival projection, cashflow inflation, market-consistent discounting, best-estimate liability calculation and deterministic sensitivity analysis.

2 Data

2.1 Mortality data

The mortality basis was taken from the Human Mortality Database (HMD), using the United Kingdom total population period life table at one-year age and one-year calendar-year granularity. HMD provides harmonised mortality data designed for demographic and actuarial analysis, with life table information available by country, sex and calendar year [3, 4].

For the base model, the total both-sex table was used. This keeps the first version of the valuation simple and avoids introducing separate male and female annuitant assumptions. The latest available mortality year in the model run was 2022.

The key mortality input for the annuity calculation is the one-year death probability at each age. This is used to calculate survival probabilities from age 65 onward, which then determine the probability-weighted annuity cashflows.

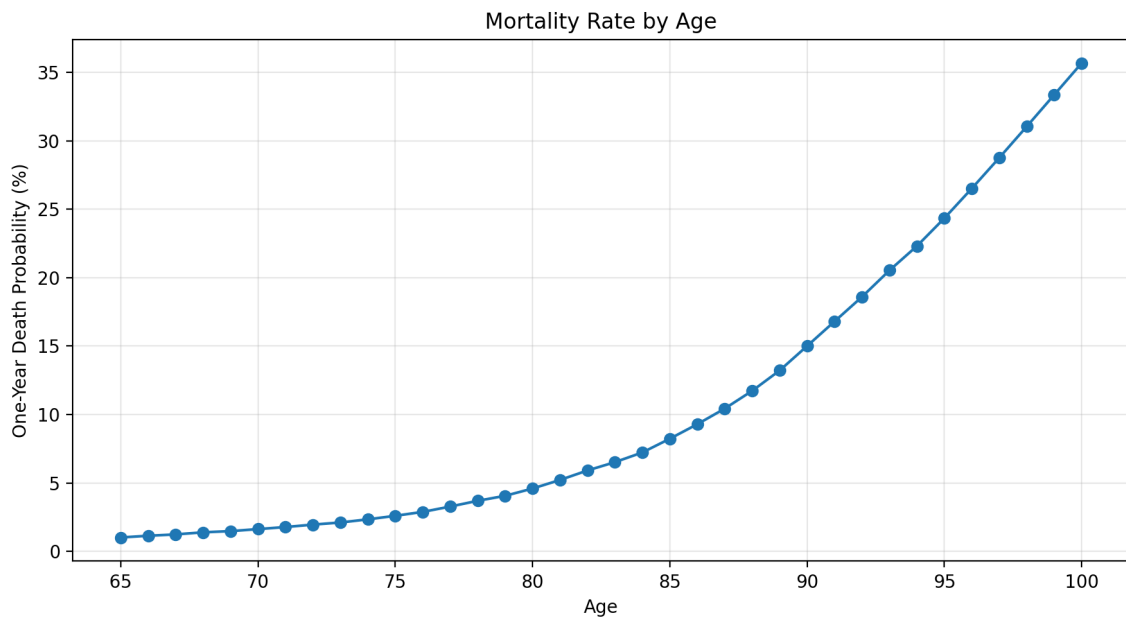


Figure 1: One-year death probability by age under the HMD mortality basis

The mortality rate increases with age, which is the key driver of the declining survival probabilities used later in the valuation. This pattern means that later annuity payments are assigned progressively lower probability weights before discounting.

2.2 Yield curve data

The discount curve was taken from the Bank of England's published UK yield curve data. The project uses the nominal gilt spot curve, which provides market-implied spot rates by maturity for UK government bonds [1, 2].

The latest available nominal spot curve observation in the model run was dated 2026-05-21. These spot rates were converted into decimal form and linearly interpolated to annual projection years before discounting the projected annuity cashflows.

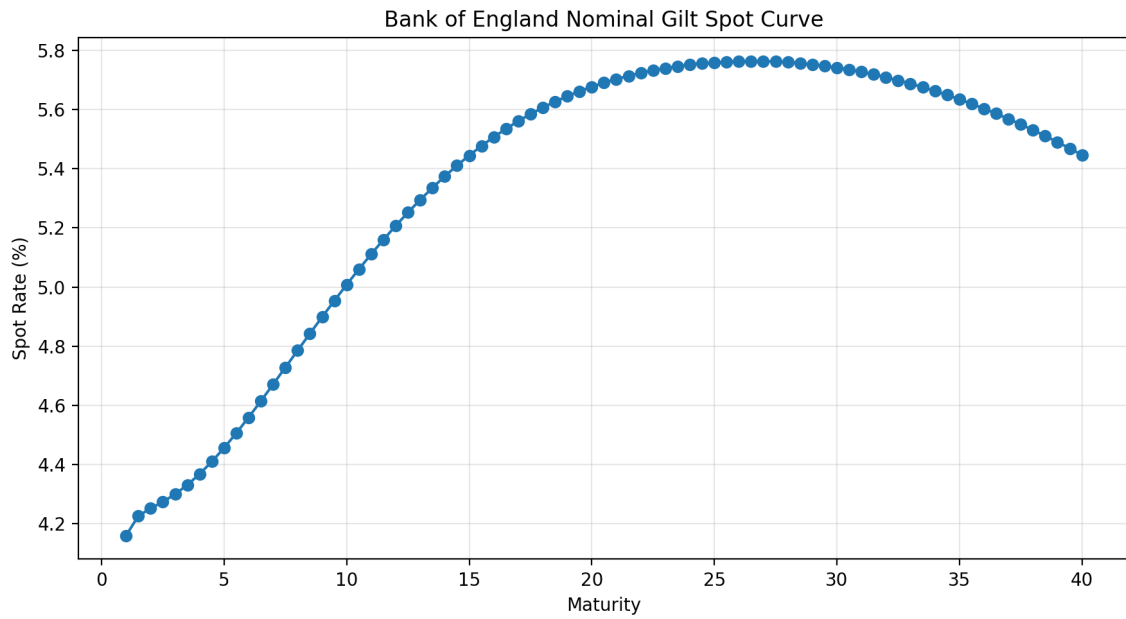


Figure 2: Bank of England nominal gilt spot curve used for discounting

The curve rises from short maturities to medium and longer maturities, then slopes down slightly at the longest maturities. This matters for annuity valuation because annuity liabilities are long-duration: payments expected further in the future are discounted using longer-maturity rates.

2.3 Data summary

Table 1: Data and valuation setup

Metric	Value
Mortality source	Human Mortality Database UK period life table
Mortality basis	Total both-sex population
Mortality year used	2022
Yield curve source	Bank of England nominal gilt spot curve
Yield curve date used	2026-05-21
Valuation age	65
Annual pension	£12,000
Annual expense	£75
Inflation assumption	2.5% per annum
Payment timing	Annual payments in arrears

3 Methodology

3.1 Survival probabilities

The annuity payments are contingent on the annuitant being alive at future payment dates. For an annuitant aged x , the probability of surviving to the payment date in year t is calculated from the one-year death probabilities:

$$p_x(t) = \prod_{j=0}^{t-1} (1 - q_{x+j}), \quad (1)$$

where q_{x+j} is the probability of death between ages $x + j$ and $x + j + 1$.

Because the model assumes payments are made in arrears, the first payment requires survival from age 65 to age 66. This is why the survival probability starts slightly below 1 rather than exactly at 1.

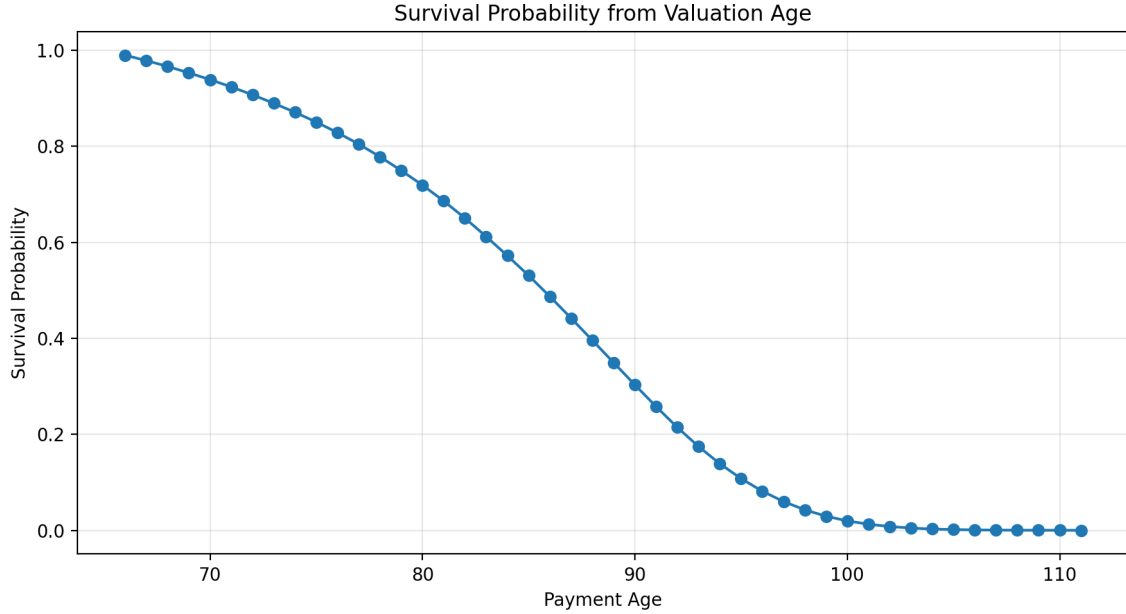


Figure 3: Survival probability from valuation age 65

The survival curve declines slowly at first and then more sharply at older ages. This pattern is consistent with increasing mortality rates at advanced ages.

3.2 Projected cashflows

The base annuity pays an annual pension of £12,000. A small annual expense allowance of £75 is also included. Both pension payments and expenses are projected forward using the inflation assumption.

The gross cashflow in projection year t is:

$$C_t = (P + E)(1 + i)^{t-1}, \quad (2)$$

where P is the annual pension payment, E is the annual expense assumption and i is the annual inflation assumption.

The survival-weighted expected cashflow is then:

$$ECF_t = p_x(t)C_t. \quad (3)$$

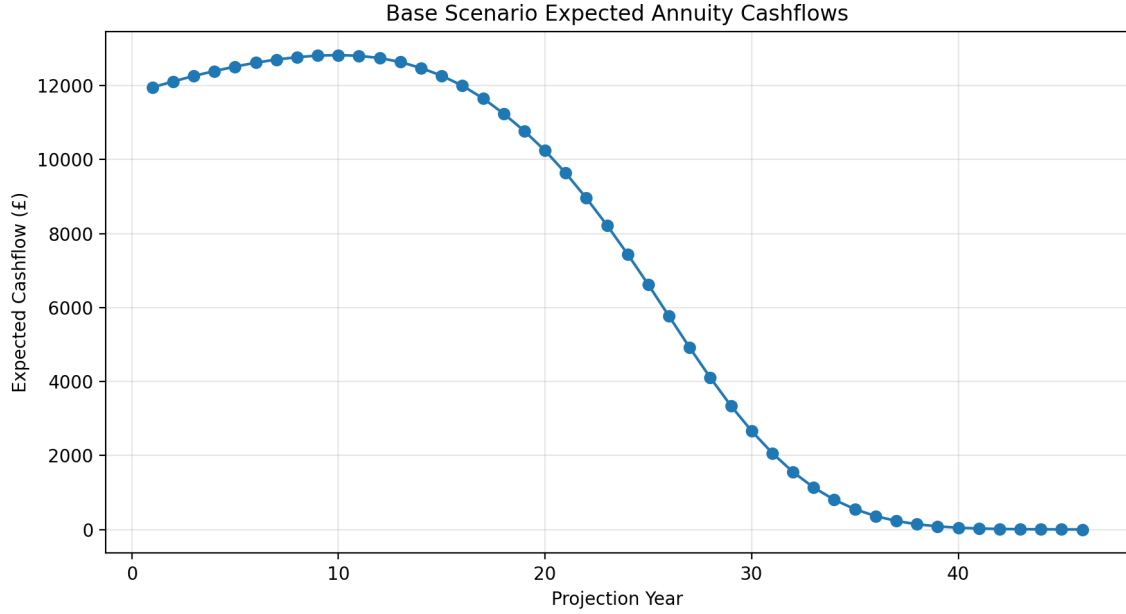


Figure 4: Base scenario survival-weighted annuity cashflows

The expected cashflow initially rises because inflation increases the pension amount. It then declines because the survival probability falls with age. This gives the expected cashflow profile a natural peak before mortality dominates the inflation effect.

3.3 Discounting

The expected cashflows are discounted using the nominal gilt spot curve. Since the curve is observed at a grid of maturities, rates are linearly interpolated to annual projection years. The Bank of England describes spot rates as rates applying from today to a given maturity, which makes them suitable for discounting projected cashflows by maturity [2].

For projection year t , the discount factor is:

$$v_t = \frac{1}{(1 + s_t)^t}, \quad (4)$$

where s_t is the interpolated nominal spot rate for maturity t .

The best-estimate liability is calculated as the sum of discounted expected cashflows:

$$BEL = \sum_{t=1}^T ECF_t v_t. \quad (5)$$

This gives a present value of the expected annuity payments and expenses under the selected mortality, inflation and discount-rate assumptions. The use of discounted expected future cashflows

is consistent with the general best-estimate liability concept used in insurance technical provision valuation [5].

4 Results and discussion

4.1 Base valuation result

The base scenario produced a best-estimate liability of:

£168,353.85

This represents the present value of future survival-weighted annuity payments and expenses for the age-65 annuitant under the base assumptions.

Table 2: Base valuation result

Metric	Value
Base BEL	£168,353.85
Annual pension	£12,000
Annual expense	£75
Inflation assumption	2.5%
Mortality year	2022
Discount curve date	2026-05-21

The BEL is substantially larger than one year's pension payment because the liability covers the expected value of many future payments. However, the present value is limited by mortality and discounting, especially at older ages where survival probabilities become small.

4.2 Sensitivity analysis

The sensitivity analysis tests how the BEL changes under deterministic stresses to key assumptions. The scenarios cover interest rates, longevity, mortality, inflation and expenses.

Table 3: BEL sensitivity summary

Scenario	BEL (£)	Impact vs base (£)	Impact vs base (%)
Base	168,354	0	0.00%
Rates +100bps	153,216	-15,138	-8.99%
Rates -100bps	186,001	17,647	10.48%
Longevity improvement: q_x -10%	173,414	5,060	3.01%
Mortality worsening: q_x +10%	163,744	-4,610	-2.74%
Inflation +100bps	184,535	16,181	9.61%
Inflation -100bps	154,203	-14,151	-8.41%
Expenses +10%	168,458	105	0.06%

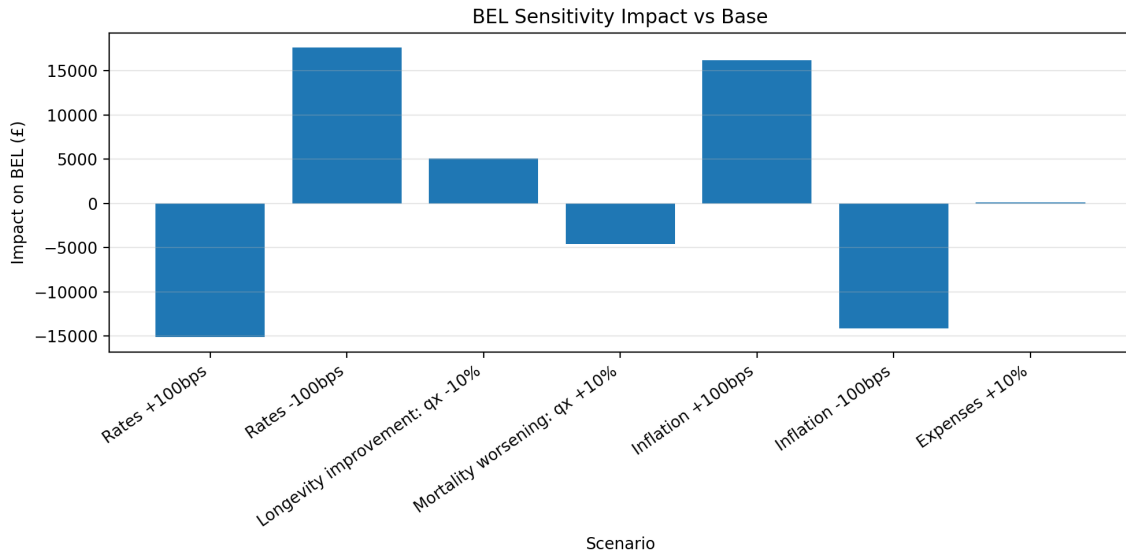


Figure 5: BEL sensitivity impact relative to the base scenario

4.3 Interpretation

The sensitivity results are consistent with the expected mechanics of annuity liability valuation.

A 100bps increase in discount rates reduces the BEL from £168,354 to £153,216. This happens because future payments are discounted more heavily. A 100bps decrease in discount rates has the opposite effect, increasing the BEL to £186,001. The rate-down impact is slightly larger in absolute terms than the rate-up impact because of the convexity of present values.

Inflation is also a major driver of the valuation. Increasing the inflation assumption by 100bps raises the BEL by £16,181, while reducing inflation by 100bps lowers it by £14,151. This is expected because pension and expense cashflows are projected with inflation.

The longevity stress also behaves as expected. Reducing death probabilities by 10% increases the BEL by £5,060 because the annuitant is expected to survive for longer and receive more payments. Increasing death probabilities by 10% reduces the BEL by £4,610.

The expense stress has a very small impact because the assumed annual expense is only £75 compared with a £12,000 pension. A 10% increase in expenses therefore adds only £7.50 per projected year before survival weighting and discounting.

Overall, the model is most sensitive to discount rates and inflation, with longevity risk also contributing a meaningful but smaller effect.

4.4 Limitations and extensions

The model is deliberately scoped to isolate the core valuation mechanics. The main limitations are:

- **Period mortality:** the model uses a period life table and does not explicitly project future mortality improvements.

- **Simplified longevity stress:** the longevity scenario applies a flat reduction to one-year death probabilities. A production model would normally use a structured mortality-improvement basis, with age- and cohort-dependent improvement assumptions.
- **Single annuitant:** the model values one representative policyholder rather than a portfolio of annuitants with different ages, pension amounts and benefit structures.
- **No sex split:** the base version uses the total both-sex table rather than separate male and female mortality. A real annuity portfolio would normally reflect the sex mix of the book, since lower mortality assumptions increase annuity liabilities.
- **Terminal age treatment:** the projection is capped at age 110, with the terminal open-age group treated as the end of the liability projection. This has limited impact on the base valuation because survival probabilities are already very small at the oldest ages.
- **Simplified inflation treatment:** pension payments and expenses are both projected using a constant inflation assumption. In practice, expense assumptions could be fixed, inflation-linked, or split into fixed and variable components.
- **Deterministic sensitivities:** the stress tests are deterministic and do not model a distribution of possible outcomes.
- **No guarantee period or spouse benefits:** the annuity is modelled as a simplified single-life annuity only.
- **No risk margin:** the output is a best-estimate style liability rather than a full regulatory technical provision.

These modelling choices keep the framework transparent, but they mean the output should be read as a best-estimate style demonstration rather than a production actuarial valuation. The model could be extended by:

- adding separate male and female mortality tables;
- valuing a portfolio of annuitants with different ages, pension amounts and benefit structures;
- incorporating structured mortality improvement assumptions;
- comparing nominal discounting with a real-rate and inflation-linked approach;
- adding stochastic interest-rate or longevity scenarios;
- including guarantee periods, spouse benefits or escalation rules.

The most natural next extension would be a small portfolio version, since that would make the output closer to a reserving or actuarial reporting exercise.

5 Conclusion

This project implemented a transparent annuity liability valuation model using Python and Excel. Mortality assumptions were based on the HMD UK total both-sex period life table, while discounting used the Bank of England nominal gilt spot curve.

The base BEL was estimated at £168,353.85 for a simplified age-65 annuity paying £12,000 per year plus expenses. Sensitivity testing showed that the liability is most exposed to discount rates and inflation, with longevity risk also affecting the valuation in the expected direction.

The project demonstrates the core workflow behind actuarial liability valuation: clean mortality and curve data, project survival-weighted cashflows, discount expected payments, calculate a best-estimate liability and assess assumption sensitivity.

References

- [1] Bank of England. *Yield curves*. Available at: <https://www.bankofengland.co.uk/statistics/yield-curves>
- [2] Bank of England. *Yield curve terminology and concepts*. Available at: <https://www.bankofengland.co.uk/statistics/yield-curves/terminology-and-concepts>
- [3] Human Mortality Database. *Human Mortality Database*. University of California, Berkeley (USA), and Max Planck Institute for Demographic Research (Germany). Available at: <https://www.mortality.org/>
- [4] Human Mortality Database. *Methods Protocol, Version 6*. Available at: <https://www.mortality.org/File/GetDocument/Public/Docs/MethodsProtocolV6.pdf>
- [5] European Insurance and Occupational Pensions Authority. *Guidelines on valuation of technical provisions*. Available at: https://www.eiopa.europa.eu/document/download/d362668c-e692-4059-b07f-2660d29a3ba9_en